

ARAVIND RAM

CLOUD INFRASTRUCTURE ENGINEER

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PROFFESIONAL SUMMARY

Database Reliability / Platform Engineer with 2+ years building and operating self-hosted database platforms at scale — ClickHouse (60 TB+, 200+ servers) and MongoDB on Kubernetes (EKS) and AWS. Authored a production Go Kubernetes operator, led a Percona → in-house MongoDB operator migration, and executed a 16 TB DocumentDB → MongoDB migration with zero data loss. Owns sharding, backup/PITR, disaster recovery, upgrades, and query performance — improved SELECT performance ~40% and cut CPU throttling ~90%.

SKILLS

- Databases : ClickHouse, MongoDB, Amazon DocumentDB, ZooKeeper; sharding, replication, PITR, backup/restore, TTL tiered storage, query optimization, load testing & benchmarking
- DB Platform & Operators: Go Kubernetes operators (Percona PSMDB → in-house CSMDDB), reconciliation logic,
- Operator SDK, CRDs, ClickHouse Operator, lifecycle management
- Reliability: Cross-region disaster recovery, hourly backups, capacity planning, RCA, 24/7 on-call, SOP authoring
- Kubernetes & IaC: EKS, Helm, ArgoCD, Karpenter, Terraform, Docker
- AWS: S3 (+ Cross-Region Replication), EC2, EKS, IAM/IRSA, EBS, Lambda, Route53
- Languages: Python, Go, Bash, Java, JavaScript

EXPERIENCE

FRESHWORKS, CHENNAI

2023- PRESENT

Database Reliability Engineer

- Database Provisioning & Operations
 - Provisioned and operated ClickHouse clusters at 60 TB+ across 200+ servers and self-hosted MongoDB, standardized with Terraform IaC and ArgoCD GitOps
 - Automated tenant provisioning with Python-generated manifests (ClickHouse, ZooKeeper, Envoy) into ArgoCD, cutting onboarding from days to minutes (~95% faster)
 - Owned database lifecycle and upgrades — compatibility review, validation testing, SOPs, coordinated production rollouts with minimal risk
- Large-Scale Migration
 - Migrated 16 TB of data from Amazon DocumentDB to self-hosted MongoDB, validating compatibility and data integrity with a zero-loss cutover
 - Built a remote migration framework using ClickHouse Remote Functions for onboarding external deployments, with automated schema creation and Route53 cutover
- Backup, PITR & Disaster Recovery
 - Designed cross-region DR — S3 Cross-Region Replication, hourly backups, snapshot-based recovery, and automated restore workflows
 - Implemented PITR for critical production tenants, strengthening business continuity

- Performance & Storage Engineering
 - Improved ClickHouse SELECT query performance ~40% by evaluating and adopting Lazy Materialization
 - Reduced CPU throttling ~90% via Kubernetes CPU Manager static allocation; conducted insert/query benchmarking to de-risk version upgrades and set capacity baselines
 - Implemented hot/cold tiered storage (EBS → S3 via TTL policies + local EBS cache), cutting storage cost while preserving query performance
- Operator Development
 - Engineered a production Kubernetes operator in Go, leading the full migration from the Percona MongoDB Operator (PSMDB) to an in-house Community MongoDB Operator (CSMDB)
 - Rewrote reconciliation logic for sharding management, backup workflows, Point-In-Time Recovery (PITR), and resource lifecycle to support internal use cases not covered by Percona
 - Built integration and E2E test coverage with Jenkins CI pipelines for operator builds and releases

PROJECTS

- Infrastructure Cost Dashboard — AWS Cost Explorer + Lambda + custom dashboarding for per-tenant spend
- ClickHouse MCP Server — AI-assisted infra operational tooling
- MongoDB Migration — 16 TB DocumentDB → self-hosted MongoDB on Kubernetes

EDUCATION

Bachelor of Science (BS) Degree in Data Science and Applications

Indian Institute of technology, Madras 2023 - 2027

COURSEWORK

CKAD: Certified Kubernetes Application Developer

LANGUAGES

Tamil - Native

English - Fluent